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Centre Number	Candidate Number	Name

CAMBRIDGE INTERNATIONAL EXAMINATIONS
International General Certificate of Secondary Education

PHYSICS **0625/06**

Paper 6 Alternative to Practical

October/November 2003

1 hour

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

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1	
2	
3	
4	
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Total	

This document consists of **11** printed pages and **1** blank page.

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1 **(a)** A student was asked to make some measurements of the test-tube shown in F
He was given a 1 m length of thin string and a metre rule and instructed to deter
as accurately as possible the circumference of the tube. The student used the rule t
measure the diameter d of the tube and then calculated the circumference c using the
equation

$$c = \pi d.$$

Describe how the student could have obtained a more accurate result with the apparatus given. You may draw on Fig. 1.1.



Fig. 1.1

.....

.....

..... [2]

(b) **(i)** On Fig. 1.2, show where you would place two small rectangular blocks of wood to help you make an accurate measurement of the overall length of the test-tube.



Fig. 1.2

(ii) The test-tube is shown actual size in Fig. 1.2. Use your rule to measure the length l of the test-tube.

l = ...

[2]

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(c) Using another test-tube, the student obtained these readings.

$l = 14.5\,\text{cm}$

$c = 5.3\,\text{cm}$

Calculate the approximate external volume V of the test-tube using the student's readings and the equation

$$V = \frac{c^2 l}{4\pi}.$$

$V =$ ..... [2]

(d) The equation used in **(c)** assumes the test-tube to be a cylinder with flat ends. It does not allow for the rounded end of the test-tube.

(i) Estimate the volume V_m of the 'missing' part of the cylinder shown shaded in Fig. 1.3.



Fig. 1.3

$V_m =$ .....

(ii) Using your values for V and V_m, calculate the actual external volume V_a of the test-tube.

$V_a =$ .....

[2]

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2 Fig. 2.1 shows a ray tracing sheet obtained by a student carrying out a reflection experiment using pins and a plane mirror.

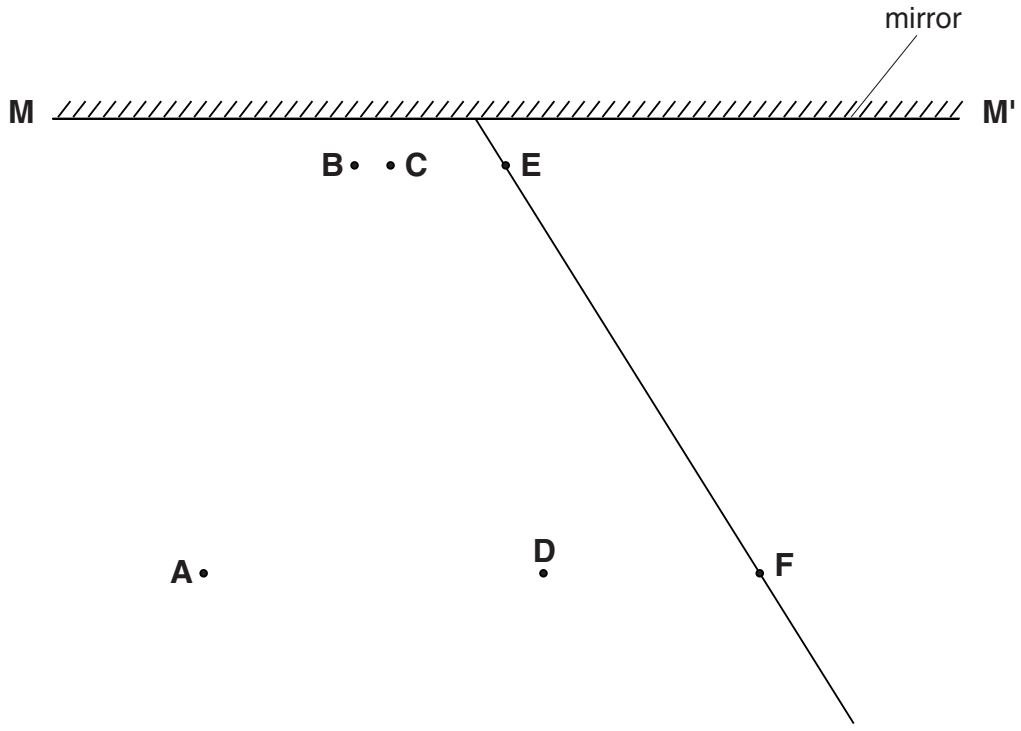


Fig. 2.1

The student looks into the mirror **MM'** and views the images of pins **A** and **B**. He then places pins **C** and **D** so that pins **C** and **D** and the images of pins **A** and **B** appear to be in line.

(a) On Fig. 2.1,

(i) draw the incident ray in this experiment,

(ii) draw the reflected ray,

(iii) draw a normal at the point where the incident ray meets the mirror,

(iv) measure the angle of incidence *i*.

i = ...

[2]

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(b) The student moves pin **B** and then repeats the experiment, obtaining the reflec
EF.

(i) On Fig. 2.1, continue the lines **CD** and **EF** behind the mirror to find the point where they meet. Label this point **X**.

(ii) Draw the line **AX**. Label with the letter **Y** the point where line **AX** crosses the mirror **MM'**.

(iii) Use your rule to measure the distances **AY** and **YX**.

AY = ...

YX = ...

[2]

(c) According to theory, **AY** = **YX**. Suggest why, in spite of very careful work, the student's values may have been slightly different.

.....

..... [1]

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3 **(a)** Fig. 3.1 shows the scale of an ammeter. Draw the position of the pointer wh ammeter reading is 0.35 A.

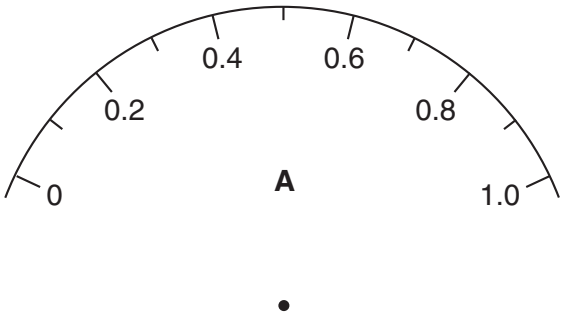


Fig. 3.1

(b) The ammeter was used in the circuit shown in Fig. 3.2 to investigate the current in a lamp.

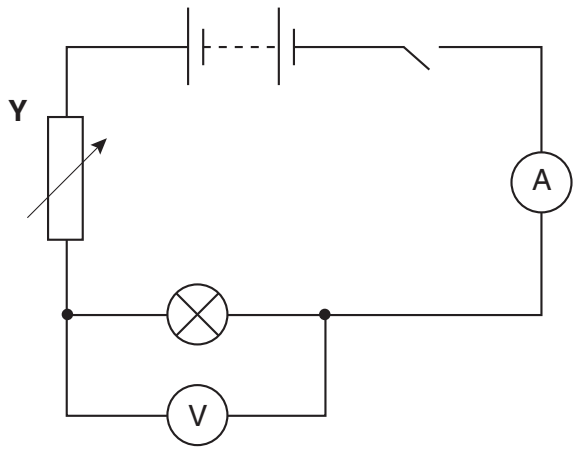


Fig. 3.2

(i) Name the component labelled **Y**.

.....

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(ii) The table shows the current I in the lamp for different values of the p.d. V the lamp.

V/	I/	R/
1.9	0.31	
1.5	0.26	
0.8	0.20	

1. Calculate the values for the resistance R of the lamp, using the equation

$$R = \frac{V}{I}.$$

Write your answers in the table.

2. Complete the column headings in the table.

(iii) Suggest how the value of V could be varied.

.....

.....

[8]

(c) Fig. 3.3 shows a power source connected to three resistors labelled **X**, **Y** and **Z**.

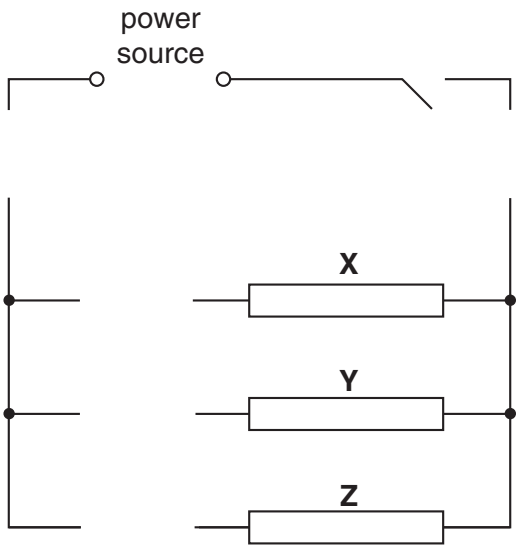


Fig. 3.3

Complete the diagram to show

(i) a voltmeter connected to measure the voltage across the resistors,

(ii) an ammeter connected to measure the current in resistor **X** only,

(iii) connecting wires to complete the circuit.

[3]

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4 The IGCSE class was studying the acceleration *a* of a toy truck that was pulled a track by a force *F*. The arrangement is shown in Fig. 4.1.

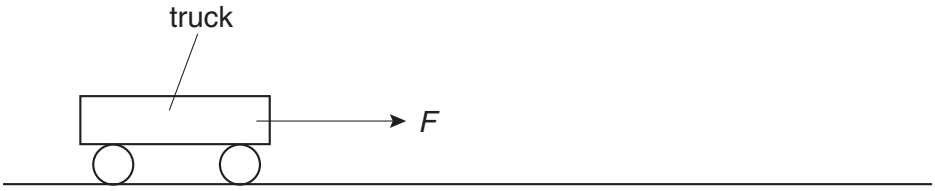
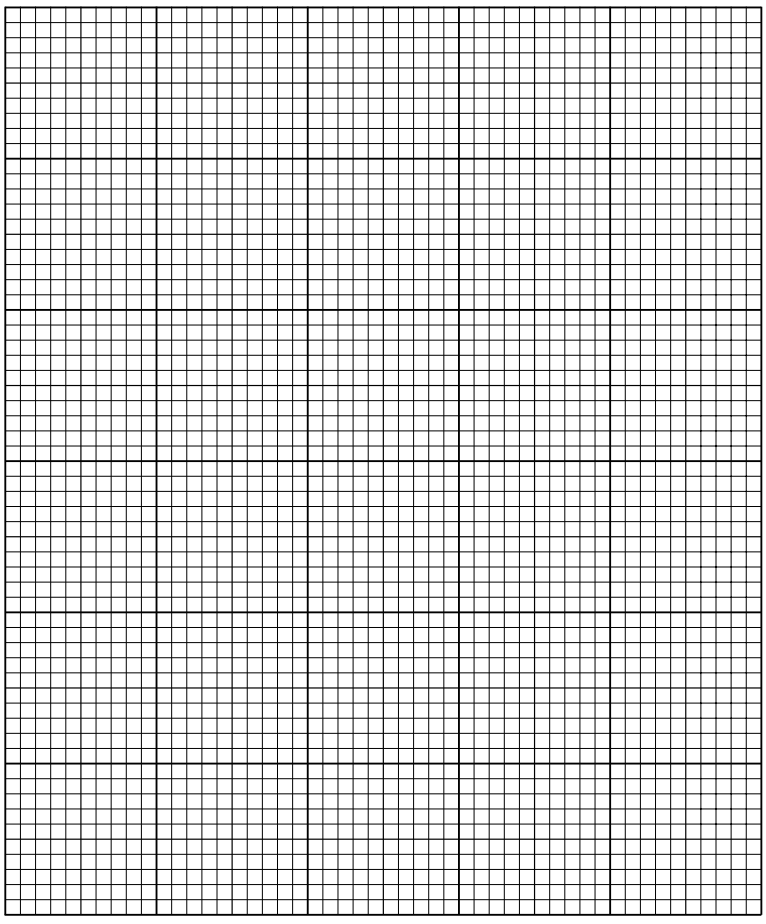


Fig. 4.1

The results obtained are shown in the table.

F/N	a/(m/s^2)
0.5	0.35
1.0	0.72
1.5	1.02
2.0	1.44
2.5	1.74

(a) Plot a graph of F/N (y-axis) against a/(m/s^2) (x-axis). Draw the line of best fit through your points. [6]



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(b) Theory suggests that the relationship between force and acceleration is given by the equation

$$F = ma,$$

where m is the mass of the truck.

The gradient of the graph is equal to the mass of the truck.

From the graph, determine the mass m of the truck. Show clearly how you obtained the necessary information.

m = ... [4]

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5 A student wants to find out which of the three materials is the best thermal insulator.

The student's apparatus at the beginning of each test is as shown in Fig. 5.1.

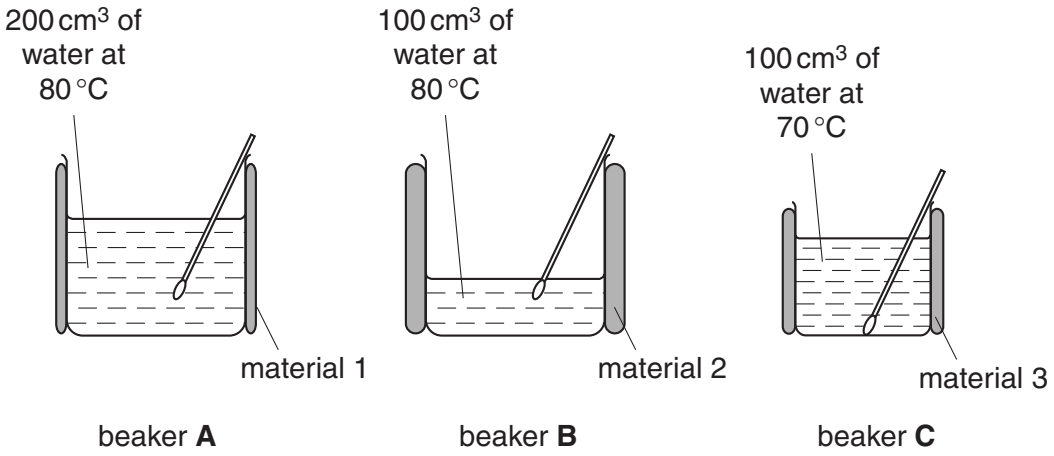


Fig. 5.1

Each beaker is surrounded by a different insulating material. The water is allowed to cool and the temperatures are recorded at different times. The student is unable to write a correct conclusion because the variables have not been controlled.

(a) Study Fig. 5.1 and then state **two** ways in which you would improve the control of variables.

1. .....

.....

2. .....

..... [2]

(b) State the reading shown on the thermometer shown in Fig. 5.2.

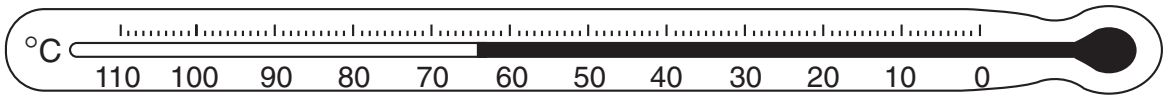


Fig. 5.2

temperature reading ..... [1]

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(c) The graph of Fig. 5.3 shows the results obtained by the student.

The graph lines **A**, **B** and **C** correspond to the beakers **A**, **B** and **C**.

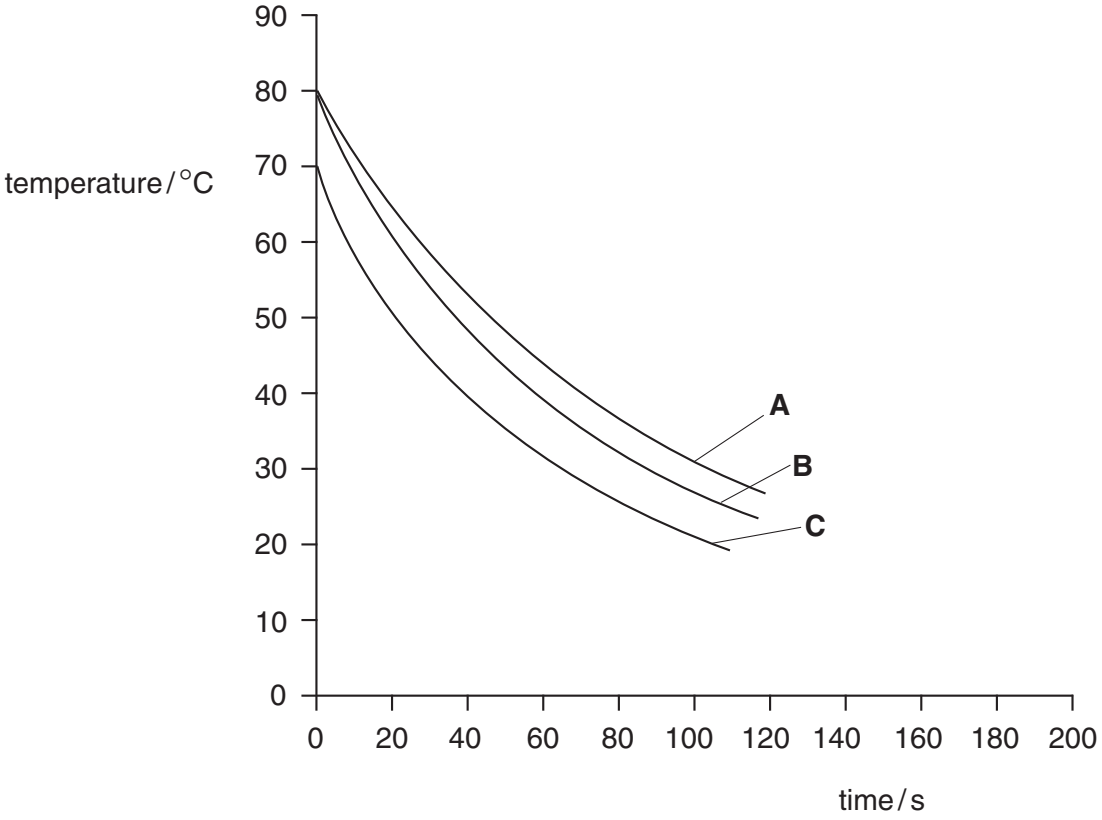


Fig. 5.3

Which beaker cools the most quickly in the first 60 s?

..... [1]

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